

HAPS FSO PROJECT

From OOK Transmission to Optical Jamming, Receiver Blinding and the ML Transition

Coverage: Baseline → EXP 11

Simulation Platform: MATLAB

System: HAPS-to-Ground Free-Space Optical Communication

Prepared from the actual project source archive and recorded experiment outputs.

Document Purpose and Reading Guide

This document is the master technical record of the HAPS FSO project through Experiment 11. It explains the system from first principles, maps every major MATLAB module to the underlying physics, documents the experiments and reconstructs the central results and patterns observed during development.

The primary source is the uploaded MATLAB project archive. Numerical plots for EXP 09–11 are reconstructed from recorded experiment outputs. Representative baseline turbulence values come from the baseline consolidated documentation contained in the project archive.

How to read this document

Reader Category	Recommended Reading Path
Beginner	Sections 1 → 7 → 10 → 14 → 20
MATLAB reviewer	Sections 3 → 13 and Appendix A
Research reviewer	Sections 14 → 24
Future ML developer	Sections 20 → 25

Project status at the end of this document

- **Baseline Model:** Legitimate HAPS FSO link is modularized and validated.
- **Jammer Implementation:** Separate constant and random optical jammer models are operational.
- **Power Sweeps:** Revealed non-monotonic BER curves and blinding transitions.
- **Validation:** Random intermittent jamming validated against activity probability sweeps.
- **Comparison:** EXP 11 contrasted constant vs. random jammers under unified power budgets.
- **Future Phase:** Transition to ML-based detection, classification, and mitigation.

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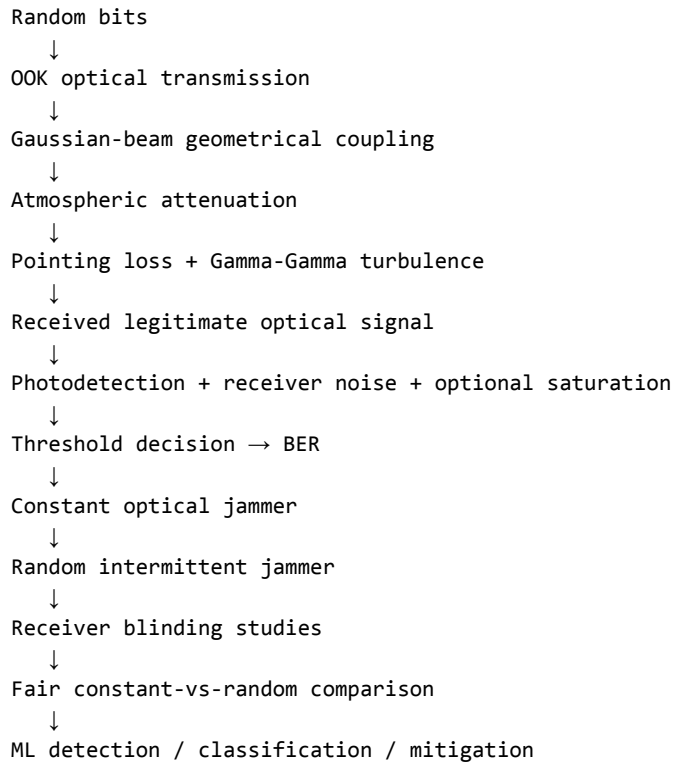
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1. Project Overview and Research Story

The project studies a HAPS-to-ground free-space optical (FSO) communication link using intensity modulation/direct detection. A binary source is converted into an on-off optical waveform, propagated through deterministic and random atmospheric effects, converted into receiver current, and detected with a fixed threshold. After the legitimate baseline is established, an adversarial optical source is added to study jamming and receiver blinding.

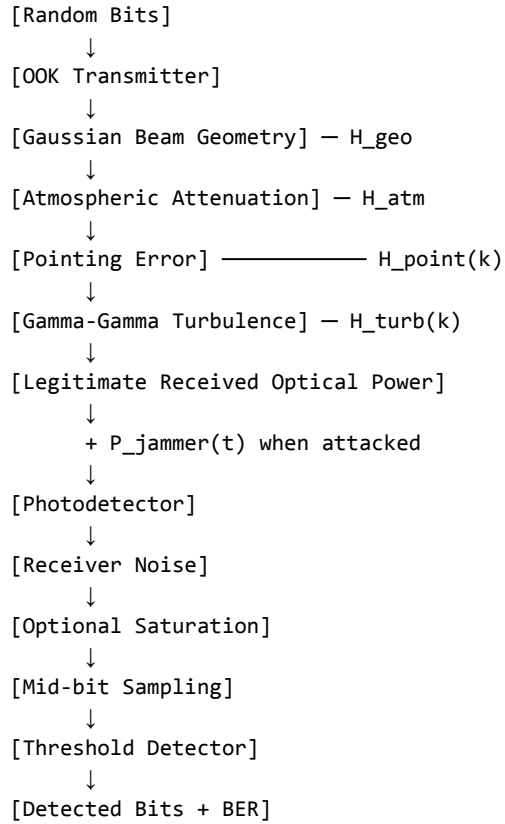
Research progression



Core questions

- How do geometry, attenuation, pointing error and turbulence affect an OOK HAPS FSO link?
- How does constant optical jamming alter BER and error type?
- Why can small jammer power temporarily improve BER in this fixed-threshold model?
- How does intermittent jamming differ from continuous jamming?
- How does receiver saturation create a blinding regime?
- Under the same average power budget, which temporal attack structure is more damaging?
- Which receiver signatures should feed the next ML phase?

2. High-Level End-to-End Architecture



$$P_{rx,legit} = P_{tx} \cdot H_{geo} \cdot H_{atm} \cdot H_{point} \cdot H_{turb}$$

$$P_{rx,total} = P_{rx,legit} + P_{jammer}$$

The legitimate channel is multiplicative. The adversarial jammer is additive at the optical input before photodetection.

3. Project Repository and Software Architecture

```

HAPS_FSO_PROJECT/
├── baseline_model/
│   ├── defaultParameters.m
│   ├── main_baseline.m
│   ├── runBaseline.m
│   └── MainModels/
│       ├── generateOOK.m
│       ├── geometryModel.m
│       ├── atmosphericAttenuation.m
│       ├── pointingErrorModel.m
│       ├── gammaGammaTurbulence.m
│       └── receiverModel.m
├── baseline with jammer/
│   ├── jammerModel.m
│   └── randomJammerModel.m
├── experiments/
│   ├── EXP_BASE_01_distanceSweep.m
│   ├── EXP_BASE_02_turbulenceSweep.m
│   ├── EXP_BASE_03_pointingSweep.m
│   ├── EXP_BASE_04_channelRealization.m
│   ├── EXP_05_constantJammer.m
│   ├── EXP_06_jammerPowerSweep.m
│   ├── EXP_07_randomJammerValidation.m
│   ├── EXP_08_randomJammerSweep.m
│   ├── EXP_09_constantJammerBlindingAttack.m
│   ├── EXP_10_randomJammerBlindingAttack.m
│   └── EXP_11_fairJammerComparison.m
└── tests/
    ├── test01_perfectChannel.m
    └── test02_receiverNoise.m

```

System Layer	Primary Responsibility
Parameters	Nominal simulation settings.
Transmitter	Bits, OOK symbols and waveform.
Propagation	Geometry and atmospheric attenuation.
Random channel	Pointing and turbulence.
Receiver	Photodetection, noise, saturation, sampling and decision.
System engine	End-to-end legitimate simulation and diagnostics.
Attack layer	Constant and random jammer generation.
Experiments	Sweeps, comparisons and plots.
Tests	Sanity and sensitivity validation.

4. Default System Parameters

System Parameter	Symbol / Physical Meaning	Default Value
Number of bits	Nbits	1000
Bit rate	R _b	1×10 ⁶ bit/s
Transmit power	P _t	1 W
Samples/bit	samplesPerBit	20
Wavelength	λ	1550 nm
Divergence half-angle	θ	0.01 mrad
Receiver diameter	D _{Rx}	0.2 m
Responsivity	R	0.8 A/W
Distance	L	20 km
Atmospheric attenuation	α	0.1 dB/km
Pointing std	σ _{point}	0.02 m
Turbulence	C _n ²	1×10 ⁻¹⁶
Noise std	σ _{noise}	0.005 A
Saturation default	enableSaturation	false
Random seed	rngSeed	1

$$T_b = 1/R_b = 1 \mu s$$

$$F_s = R_b \cdot \text{samplesPerBit} = 20 \text{ MHz}$$

The 20 samples per bit provide numerical waveform resolution. In the present implementation, pointing and turbulence are generated once per bit and repeated over that bit's samples.

5. Binary Information and OOK Transmission

5.1 Random bit source

```
bits = randi([0 1], 1, Nbits);
```

$$\mathbf{b}_k \in \{0, 1\}$$

5.2 On-Off Keying

```
ookSymbols = Pt * bits;
```

$$P_{tx}(k) = 0 \text{ for } b_k=0, \text{ and } P_{tx}(k) = P_t \text{ for } b_k=1$$

OOK maps information directly to optical intensity: zero means no legitimate optical contribution in this simplified model; one means P_t .

5.3 Waveform expansion

```
txSignal = repelem(ookSymbols, samplesPerBit);
```

Each bit-level symbol is repeated across 20 samples so the waveform can be processed by the channel and receiver.

5.4 Time axis

```
Fs = Rb * samplesPerBit;  
t = (0:length(txSignal)-1) / Fs;
```

6. Geometrical Propagation

geometryModel.m models Gaussian-beam spreading and the fraction of beam power captured by a finite circular receiver.

$$w_0 = \lambda / (\pi\theta)$$

$$z_R = \pi w_0^2 / \lambda$$

$$w_{Rx} = w_0 \sqrt{1 + (L / z_R)^2}$$

$$H_{geo} = 1 - \exp(-2R_{Rx}^2 / w_{Rx}^2)$$

As propagation distance increases, the beam spreads. A finite receiver aperture therefore captures a changing fraction of the Gaussian optical power.

The baseline consolidated documentation recorded a representative 20 km operating point of $H_{geo} \approx 0.375821$ and $w_{Rx} \approx 0.2060$ m for the project parameterization.

7. Atmospheric Attenuation

$$\begin{aligned}L_{\text{km}} &= L / 1000; \\ H_{\text{atm}} &= 10^{(-\alpha * L_{\text{km}} / 10)};\end{aligned}$$

$$H_{\text{atm}} = 10^{-\alpha L_{\text{km}}/10}$$

This converts the dB/km attenuation coefficient into a linear optical power gain. The present model is path-averaged and deterministic.

$$H_{\text{total}} = H_{\text{geo}} \cdot H_{\text{atm}}$$

At $\alpha=0.1$ dB/km and $L=20$ km, the consolidated baseline documentation records $H_{\text{atm}} \approx 0.630957$.

8. Pointing Error

```
x_error = sigmaPoint * randn(1,Nbits);  
y_error = sigmaPoint * randn(1,Nbits);  
r_error = sqrt(x_error.^2 + y_error.^2);  
H_point = exp(-2*r_error.^2/wRx^2);
```

$$x_e, y_e \sim N(0, \sigma_{\text{point}}^2)$$

$$r_e = \sqrt{(x_e^2 + y_e^2)}$$

$$H_{\text{point}}(k) = \exp(-2r_e(k)^2 / w_{Rx}^2)$$

One random pointing loss is generated per bit and repeated across the samples of that bit. Severe misalignment reduces legitimate received power and increases the chance of 1→0 errors.

9. Gamma-Gamma Atmospheric Turbulence

gammaGammaTurbulence.m represents scintillation as a random multiplicative gain.

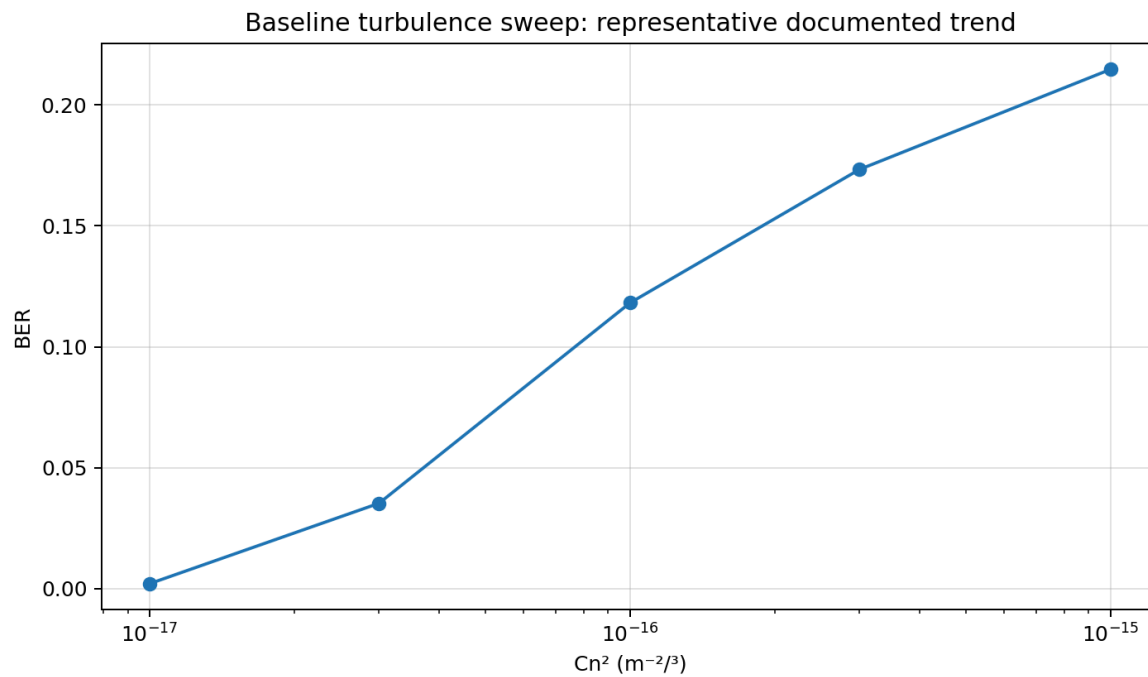
$$k = 2\pi / \lambda$$

$$\sigma_R^2 = 1.23 \cdot C_n^2 \cdot k^{7/6} \cdot L^{11/6}$$

$$H_{\text{turb}} = X \cdot Y$$

The implementation derives Gamma-Gamma parameters α_{GG} and β_{GG} from the Rytov variance, generates normalized Gamma variables X and Y, and multiplies them.

When $C_n^2=0$, the source explicitly returns $H_{\text{turb}}=1$, giving a clean no-turbulence validation condition.



Representative baseline turbulence sweep trend documented in the project's consolidated baseline report.

The representative documented values rise from BER ≈ 0.0021 at $C_n^2=1 \times 10^{-17}$ to BER ≈ 0.2147 at $C_n^2=1 \times 10^{-15}$. Exact values depend on random realizations.

10. Combined Legitimate FSO Channel

$$P_{rx}(t) = P_{tx}(t) \cdot H_{geo} \cdot H_{atm} \cdot H_{point}(t) \cdot H_{turb}(t)$$

$$P_{rx}(t) = P_{tx}(t) \cdot H_{total} \cdot H_{point}(t) \cdot H_{turb}(t)$$

```
rxOpticalSignal =  
    H_total .* H_point_signal .* H_turb_signal .* txSignal;
```

For bit 0, the legitimate optical contribution is ideally zero. For bit 1, received power fluctuates because pointing and turbulence vary.

A critical design choice is that the receiver threshold uses H_{total} rather than instantaneous $H_{total} \cdot H_{point} \cdot H_{turb}$. Consequently, severe fading can push legitimate ones below threshold.

11. Receiver: Photodetection, Noise, Saturation and Decision

11.1 Photodetection

$$I_{\text{photo}}(t) = R \cdot P_{\text{rx}}(t)$$

```
rxCurrent = R * rxOpticalSignal;
```

11.2 Receiver noise

$$n(t) \sim N(0, \sigma_{\text{noise}}^2)$$

$$I_{\text{noisy}} = I_{\text{photo}} + n$$

Noise is generated outside receiverModel so experiments can reuse the exact same realization.

11.3 Hard saturation

$$I_{\text{out}} = \min(I_{\text{noisy}}, I_{\text{sat}})$$

```
if enableSaturation
    I_sat = 0.25;
    rxCurrent = min(rxCurrent, I_sat);
end
```

This hard-clipping model is central to the blinding studies.

11.4 Mid-bit sampling

```
sampleIndices = samplesPerBit/2 : samplesPerBit : length(rxCurrent);
```

11.5 Threshold and decision

$$I_{\text{th}} = R \cdot P_t \cdot H_{\text{total}} / 2$$

$$\hat{b}_k = 1 \text{ if } I_{\text{sample},k} > I_{\text{th}}; \text{ otherwise } 0$$

11.6 BER and error types

$$\text{BER} = \text{errors} / N_{\text{bits}}$$

$$\text{Errors}_{10} = \text{count}(b=1, \hat{b}=0), \text{Errors}_{01} = \text{count}(b=0, \hat{b}=1)$$

The split between 1→0 and 0→1 errors becomes the main diagnostic tool for interpreting attack mechanisms.

12. The runBaseline Simulation Engine

runBaseline.m is intentionally the legitimate-system engine: no jammer is injected inside it.

Step	Action
1	Set random seed.
2	Generate bits and OOK waveform.
3	Compute geometry.
4	Compute atmospheric attenuation.
5	Form H_{total} .
6	Generate and expand pointing gains.
7	Generate and expand turbulence gains.
8	Apply multiplicative legitimate channel.
9	Generate receiver noise.
10	Run receiver.
11	Store diagnostics, BER, error types and channel information.

Important returned fields

- bits / detectedBits
- txSignal / rxOpticalSignal
- rxCurrent / rxSamples
- receiverNoise
- mean and standard deviation of samples for transmitted 0 and 1
- BER, threshold, bitErrors, errors10, errors01
- H_{geo} , H_{atm} , H_{total} , $H_{\text{point_bits}}$, $H_{\text{turb_bits}}$
- w_{Rx} , w_0 , z_R , α_{GG} , β_{GG} , σ_{R2}

Storing results.receiverNoise fixed an important development issue and enabled fair attacked-versus-clean comparisons.

13. Testing and Baseline Validation

Test 01 — Perfect / Ideal Channel

```
params.sigmaPoint = 0;  
params.Cn2 = 0;  
params.sigmaNoise = 0;
```

This verifies that the modulation, channel plumbing, sampling and decision logic can operate correctly when impairments are removed.

Test 02 — Receiver Noise Validation

The source uses a perfect channel and studies receiver-noise effects, with Nbits=100000 for more reliable BER statistics.

Validation philosophy

Research Question	Validation Mechanism
Does the system work ideally?	Perfect-channel test.
Does noise hurt performance?	Noise validation.
Does distance affect propagation?	Distance sweep.
Does stronger turbulence degrade the link?	Cn ² sweep.
Does misalignment degrade the link?	Pointing sweep.
Do gains actually vary?	Channel realization.

14. Baseline Experiments: EXP_BASE_01 to EXP_BASE_04

EXP_BASE_01 — Distance Sweep

The source sweeps distance from 1 km to 30 km. The intended physical trend is reduced received margin as propagation losses and beam spreading become more severe.

$L \uparrow \rightarrow$ **received signal margin generally** $\downarrow \rightarrow$ **BER eventually** $\uparrow$

EXP_BASE_02 — Turbulence Sweep

Distance and pointing are fixed while C_n^2 is varied to isolate atmospheric scintillation.

EXP_BASE_03 — Pointing Sweep

Pointing-error standard deviation is varied to isolate misalignment-induced coupling loss.

EXP_BASE_04 — Channel Realization

This experiment emphasizes that the channel contains random per-bit gains rather than a single constant number.

Together, these experiments establish the legitimate physical baseline before adversarial optical injection is introduced.

15. Threat Model: Optical Jamming

$$P_{\text{total}}(t) = P_{\text{legit}}(t) + P_{\text{jammer}}(t)$$

The attacker injects optical power at the receiver input abstraction. The receiver then converts the total optical power into electrical current.

Attack Family	Mathematical Model
Constant jammer	$P_{\text{jammer}}(t) = P_J$ continuously.
Random jammer	$J_k \sim \text{Bernoulli}(p)$; $P_{\text{jammer},k} = P_{\text{ON}} \cdot J_k$.

Attack mechanisms

- Positive optical bias can shift samples relative to threshold.
- Strong bias can cause zeros to become false ones.
- Sufficient optical power can trigger hard receiver saturation and create a blinding regime.

16. Constant Jammer and EXP 05–06

jammerModel.m

```
if ~enableJammer
    jammerSignal=zeros(1,Nsamples);
else
    jammerSignal=jammerPower*ones(1,Nsamples);
end
```

$$P_{\text{jammer}}(\mathbf{t}) = P_J$$

EXP 05 — Controlled constant-jammer comparison

The experiment explicitly aims to preserve transmitted bits, pointing realization, turbulence realization and receiver noise so that the jammer is the intended changing factor.

EXP 06 — Jammer power sweep

The sweep records BER and both error types. Development results exposed a non-monotonic curve: initially BER decreased as positive optical bias rescued faded ones; at larger power the attack created dominant 0→1 errors and BER approached 0.5.

```
Weak legitimate '1' below threshold → 1→0
Weak legitimate '1' + small jammer → may cross threshold correctly
Logical '0' + sufficiently strong jammer → crosses threshold → 0→1
```

The key lesson is that jammer power must be interpreted by mechanism, not by the simplistic rule 'more jammer always means more BER'.

17. Random Jammer and EXP 07–08

randomJammerModel.m

$$J_k \sim \text{Bernoulli}(p)$$

$$P_{\text{jammer},k} = P_{\text{ON}} \cdot J_k$$

```
jammerBits = rand(1,Nbits) < activityProbability;  
jammerPowerBits = jammerPower .* jammerBits;  
jammerSignal = repelem(jammerPowerBits,samplesPerBit);
```

The jammer chooses an ON/OFF state once per bit and remains in that state across the bit's 20 samples.

EXP 07 — Validation

A recorded 50-bit validation used requested $p=0.30$ and observed actual activity 0.38. This finite-sample deviation is expected.

$$\text{Actual activity} = \text{number of ON bits} / N_{\text{bits}}$$

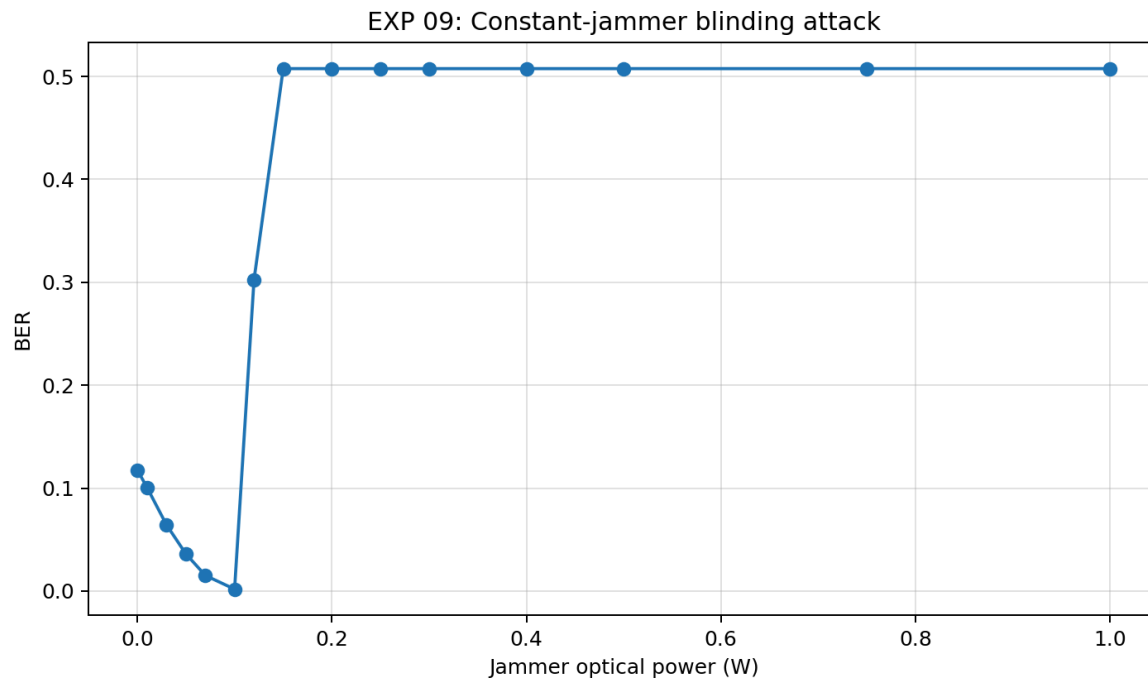
EXP 08 — Activity sweep

Activity probability is swept from 0 to 1 at fixed ON-power. The experiment demonstrates that activity is an independent attack control parameter and that the resulting BER depends on the receiver operating regime, threshold and saturation configuration.

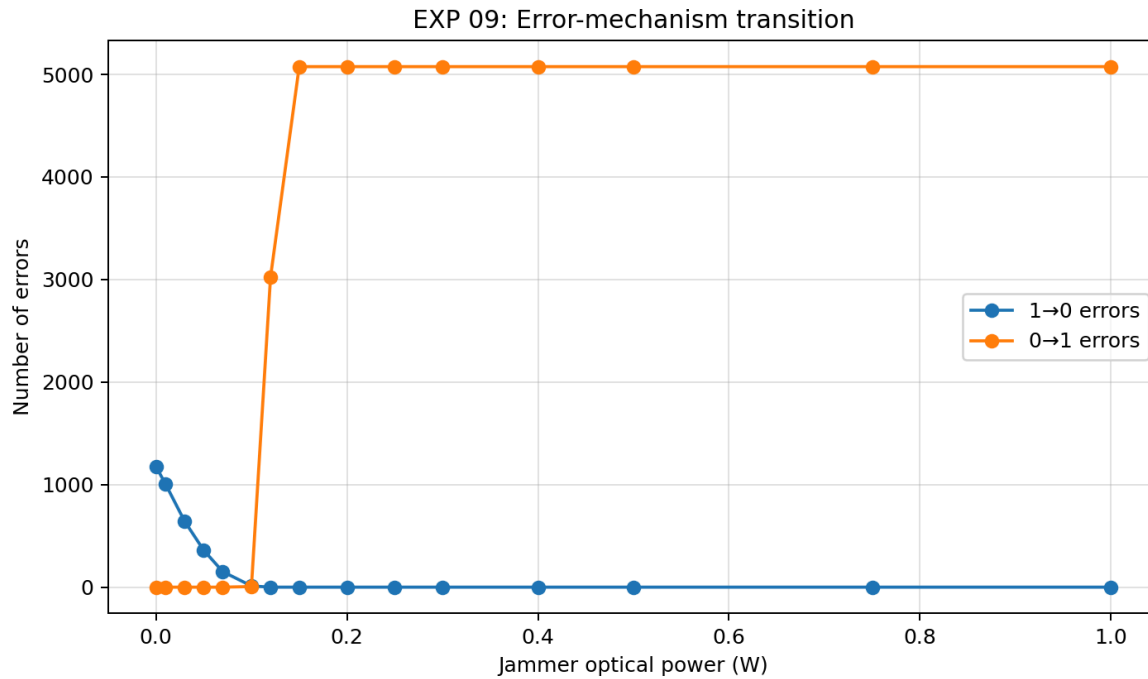
18. EXP 09 — Constant Jammer Blinding Attack

EXP 09 uses Nbits=10000, enables saturation and sweeps jammer power into the overload region.

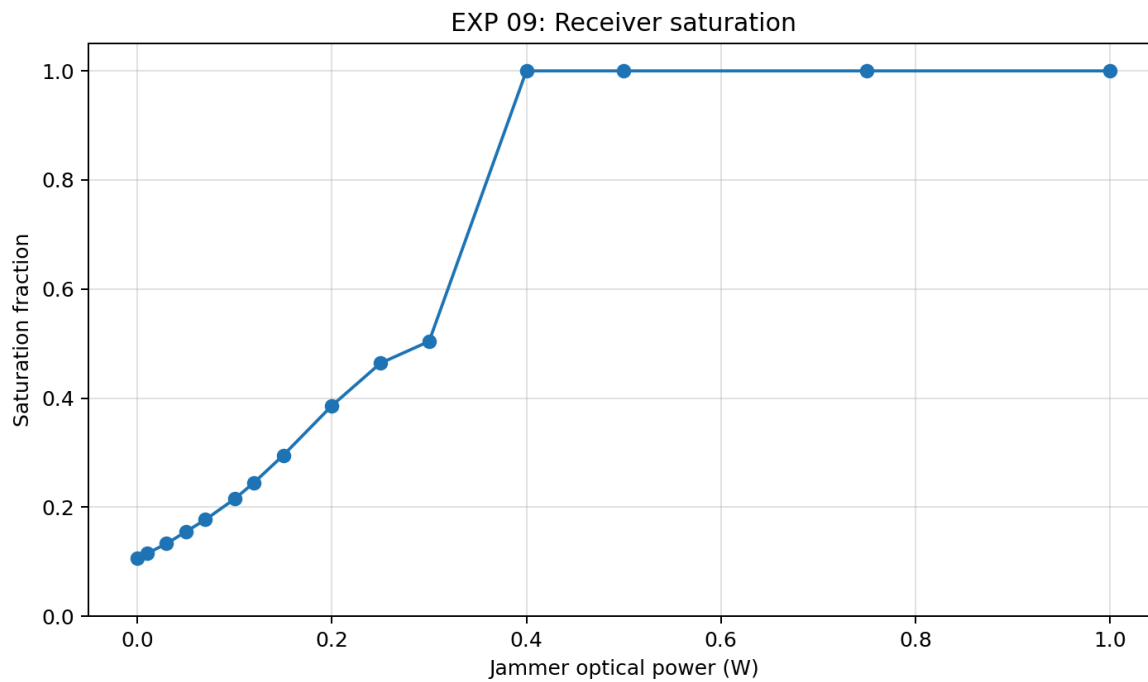
P _j (W)	Bit Error Rate (BER)	1→0 Errors	0→1 Errors	Saturation Fraction
0.000	0.1177	1177	0	0.107
0.010	0.1006	1006	0	0.115
0.030	0.0643	643	0	0.133
0.050	0.0360	360	0	0.155
0.070	0.0151	151	0	0.177
0.100	0.0018	12	6	0.215
0.120	0.3026	0	3026	0.245
0.150	0.5076	0	5076	0.295
0.200	0.5076	0	5076	0.386
0.250	0.5076	0	5076	0.464
0.300	0.5076	0	5076	0.504
0.400	0.5076	0	5076	1.000
0.500	0.5076	0	5076	1.000
0.750	0.5076	0	5076	1.000
1.000	0.5076	0	5076	1.000



EXP 09: recorded BER versus constant jammer power.



EXP 09: recorded error-mechanism transition.



EXP 09: recorded receiver saturation fraction.

Regime interpretation

Operating Region	Observed Link Behavior	Physical Mechanism
Baseline/weak	BER begins near 0.118	Mostly fading-driven 1→0 errors.
Helpful bias	BER falls to 0.0018 near 0.10 W	Bias rescues faded ones.

Transition	BER jumps at 0.12 W	0→1 errors dominate.
Blinding plateau	BER ≈0.5076	Receiver effectively outputs one for nearly all bits.
Full saturation	Sat=1 at high power	Hard clipping active for all samples.

If $\hat{b} \approx 1$ for all bits, BER $\approx$ fraction of transmitted zeros ≈ 0.5

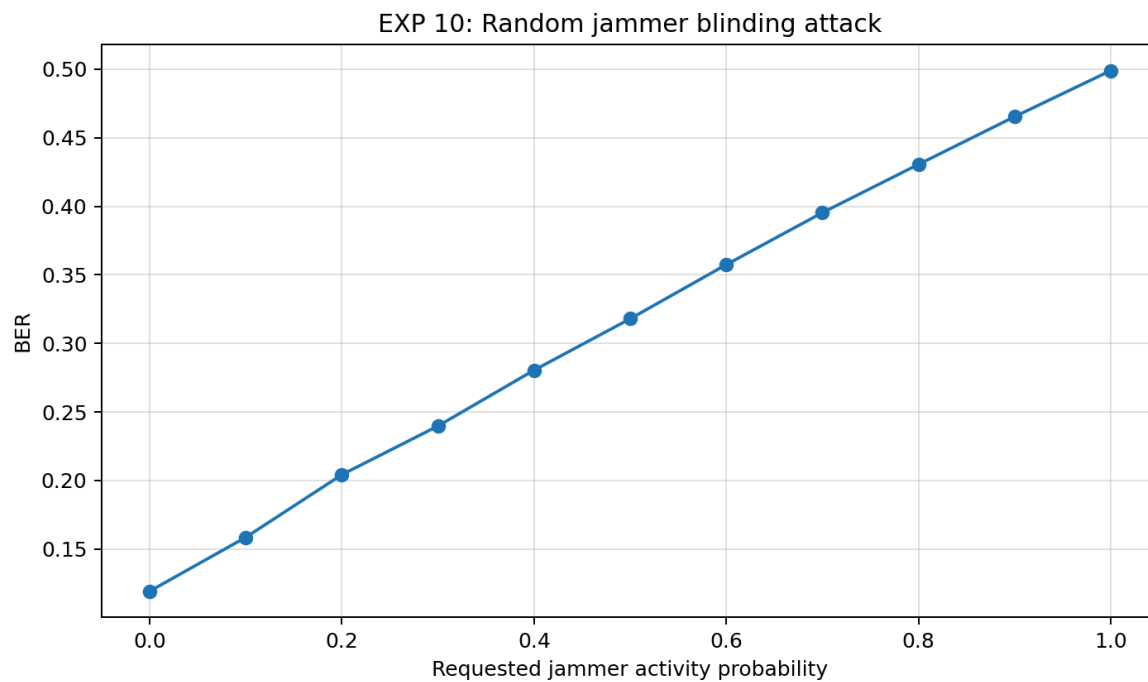
The recorded 0.5076 plateau corresponds to 5076 zero bits in the 10000-bit realization.

19. EXP 10 — Random Jammer Blinding Attack

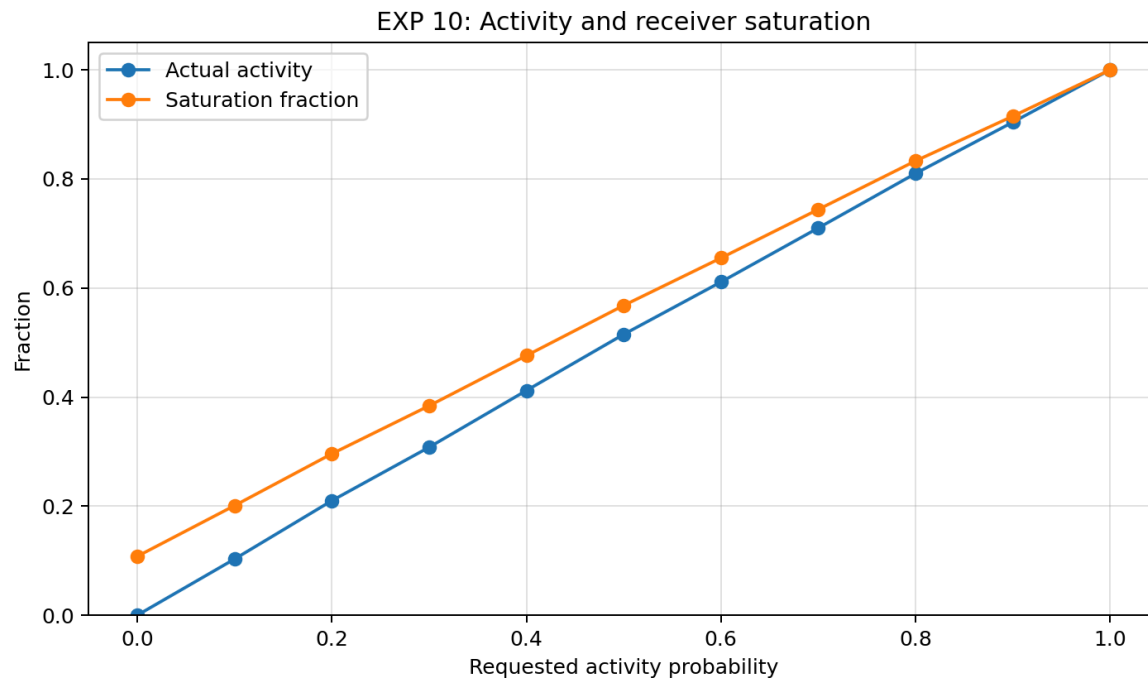
EXP 10 fixes peak jammer power at 0.4 W and sweeps activity probability. The attacked signal is formed by adding the random jammer to the legitimate baseline signal, while the stored baseline receiver noise is reused.

$$\text{Expected average jammer power} = p \cdot P_{\text{ON}}$$

p (Requested)	Actual p	BER	1→0 Errors	0→1 Errors	Saturation	Avg P _j (W)
0.0	0.000	0.1193	1193	0	0.108	0.0000
0.1	0.103	0.1586	1066	520	0.201	0.0412
0.2	0.210	0.2043	959	1084	0.296	0.0840
0.3	0.308	0.2398	835	1563	0.384	0.1234
0.4	0.412	0.2805	717	2088	0.476	0.1648
0.5	0.515	0.3180	586	2594	0.568	0.2060
0.6	0.611	0.3575	473	3102	0.655	0.2444
0.7	0.710	0.3954	368	3586	0.744	0.2839
0.8	0.810	0.4306	241	4065	0.833	0.3239
0.9	0.904	0.4654	130	4524	0.915	0.3617
1.0	1.000	0.4990	0	4990	1.000	0.4000



EXP 10: BER increases progressively with random jammer activity.



EXP 10: actual activity and saturation fraction.

As activity rises, $1 \rightarrow 0$ errors fall while $0 \rightarrow 1$ errors rise strongly. At $p=1$ the random jammer is continuously ON and the BER reaches approximately 0.499 with full saturation.

20. EXP 11 — Fair Constant vs Random Jammer Comparison

This experiment controls the average optical power budget, making it the strongest comparison experiment before the ML phase.

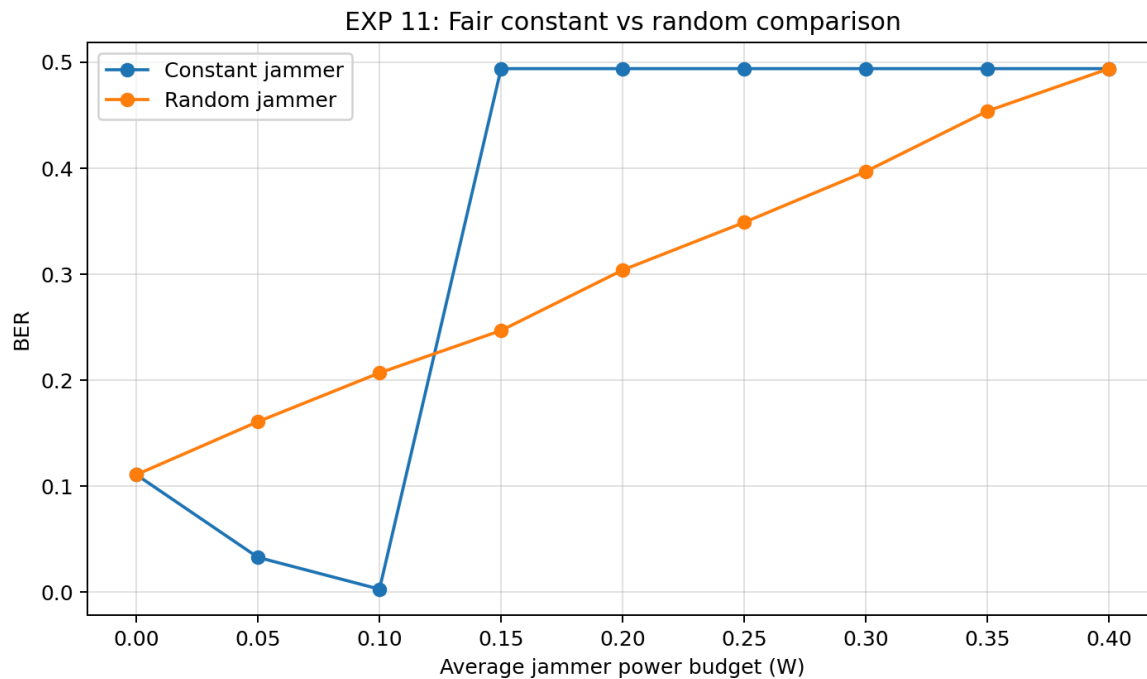
Fairness equation

$$P_{avg} = p \cdot P_{ON}$$

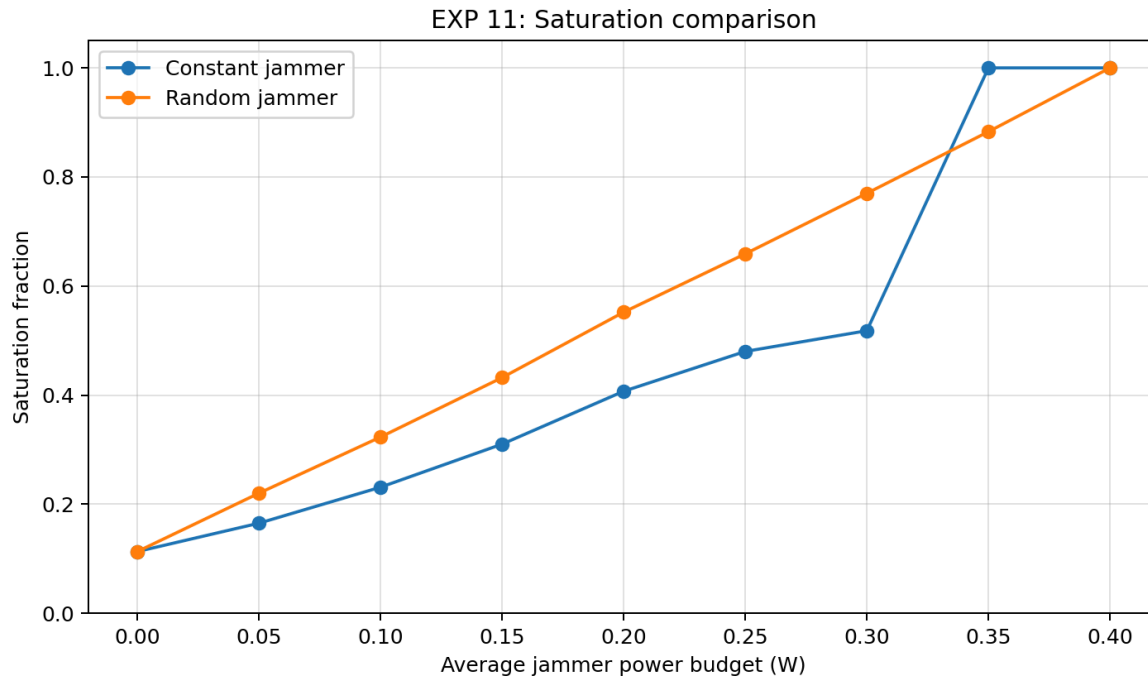
$$p = P_{avg} / P_{ON}$$

The source fixes $P_{ON}=0.4$ W. The constant jammer uses P_{avg} continuously; the random jammer uses peak power 0.4 W with activity chosen from the equation above.

Budget Pavg	Random p	Actual Random Pavg	BER (Const)	BER (Random)	Sat (Const)	Sat (Random)
0.00	0.000	0.0000	0.111	0.111	0.113	0.113
0.05	0.117	0.0468	0.033	0.161	0.165	0.220
0.10	0.239	0.0956	0.003	0.207	0.231	0.323
0.15	0.365	0.1460	0.494	0.247	0.310	0.432
0.20	0.495	0.1980	0.494	0.304	0.407	0.552
0.25	0.618	0.2472	0.494	0.349	0.480	0.659
0.30	0.740	0.2960	0.494	0.397	0.518	0.770
0.35	0.868	0.3472	0.494	0.454	1.000	0.883
0.40	1.000	0.4000	0.494	0.494	1.000	1.000



EXP 11: BER comparison under equal requested average power budget.



EXP 11: saturation comparison under equal requested average power budget.

Interpretation

At $P_{avg}=0.05$ and 0.10 W, the constant jammer sits in the helpful-bias region and produces very low BER, while the random jammer remains damaging because every ON event has a 0.4 W instantaneous amplitude. At $P_{avg}=0.15$ W the constant jammer abruptly enters its false-one regime. The random jammer degrades more gradually as a larger fraction of bits experience the strong ON-state.

This is the project's strongest evidence that average power alone does not determine attack impact. Temporal distribution and instantaneous peak power matter.

21. Cross-Experiment Patterns and Physical Interpretation

Pattern 1 — Non-monotonic BER

$P_j \uparrow$ does not necessarily imply $BER \uparrow$

Small positive bias can compensate for fading before becoming destructive.

Pattern 2 — Error asymmetry explains the curve

Weak/no jammer: mostly $1 \rightarrow 0$

Helpful bias: $1 \rightarrow 0$ decreases

Transition: $0 \rightarrow 1$ begins

Strong/blinding: almost entirely $0 \rightarrow 1$

Pattern 3 — Saturation and BER are different observables

Saturation can increase before BER reaches its maximum because BER depends on threshold crossings and symbol decisions.

Pattern 4 — Temporal attack structure matters

Constant jamming shifts every bit; random jamming alternates between normal and high-power attack states.

Pattern 5 — Blinding plateau near 0.5

An almost-always-one receiver makes every transmitted zero incorrect.

Pattern 6 — Peak power matters

EXP 11 shows that a random attack can be damaging despite a moderate average power because its instantaneous ON-power is high.

22. Reproducibility, Fairness and Debugging Lessons

Random seed control

Comparable simulations reset or fix RNG state so legitimate bits and channel realizations are reproducible.

Same legitimate realization

A jammer comparison should not simultaneously change turbulence, pointing or bits.

Same noise realization

runBaseline stores receiverNoise and attack experiments reuse it.

receiverNoise bug

The MATLAB error 'Unrecognized field name receiverNoise' was solved by storing results.receiverNoise=receiverNoise.

Finite statistics

Short validation runs can deviate substantially from requested probabilities; larger Nbits and Monte Carlo averaging improve confidence.

23. Model Assumptions and Limitations

- OOK uses 0 W for bit 0 and P_t for bit 1.
- Pointing and turbulence are independent block-fading realizations per bit.
- The turbulence process is not explicitly time-correlated across bits.
- Atmospheric attenuation is deterministic and path-averaged.
- The jammer is injected at the receiver optical-input abstraction; its own propagation path is not modeled separately.
- Receiver noise is additive Gaussian current noise.
- Saturation is hard clipping at $I_{\text{sat}}=0.25$ A.
- The threshold uses deterministic H_{total} rather than instantaneous fading estimates.
- The receiver samples once at mid-bit rather than integrating over the full bit.
- Single-run BER values retain finite-random-realization dependence.

These are appropriate simplifications for isolating causal mechanisms. A future version can increase physical realism after the ML objective is fixed.

24. Research Conclusions Through EXP 11

- The project successfully builds a modular HAPS FSO communication chain from bits to BER.
- Baseline fading primarily creates 1→0 errors under the fixed-threshold operating regime.
- A small constant optical bias can temporarily improve BER by rescuing faded ones.
- Beyond a transition point, optical injection creates dominant 0→1 errors.
- Strong optical injection plus receiver saturation produces a blinding regime with BER near 0.5.
- Random intermittent jamming is naturally controlled through a Bernoulli activity probability.
- Random and constant jammers produce different error and saturation signatures.
- Equal average jammer power does not imply equal damage because temporal structure and peak power matter.
- The signatures discovered through EXP 11 provide a principled basis for ML attack detection and mitigation.

25. Bridge to Machine Learning

The next phase should build ML on the physical signatures already discovered, rather than choosing a model first.

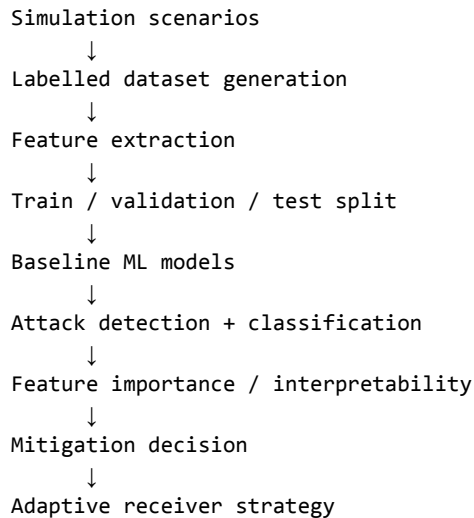
Possible tasks

ML Task	Target Output
Attack detection	Normal vs attacked
Attack classification	No attack / constant / random / blinding
Severity classification	Weak / transition / severe / blinded
Mitigation control	Adaptive receiver response

Candidate features already available

Feature Candidate	Physical Meaning
meanBit0 / meanBit1	Class-conditional received current levels.
stdBit0 / stdBit1	Class-conditional variability.
BER	Aggregate degradation.
errors10 / errors01	Mechanism asymmetry.
Saturation fraction	Receiver overload indicator.
Mean/max current	Operating level.
Threshold	Nominal decision boundary.
Channel statistics	Legitimate propagation context.

Recommended next architecture



Do not train on BER alone. BER hides the difference between fading, helpful bias, transition and blinding. The ML dataset should preserve receiver-level statistics and error asymmetry.

Appendix A. File-by-File Code Map

Module File	Functional Role
defaultParameters.m	Central nominal parameter factory.
main_baseline.m	Normal baseline entry point.
runBaseline.m	Complete legitimate simulation engine and results container.
generateOOK.m	Bits, OOK waveform and time axis.
geometryModel.m	Gaussian beam geometry and aperture coupling.
atmosphericAttenuation.m	dB/km to linear optical gain.
pointingErrorModel.m	Per-bit pointing displacement and coupling loss.
gammaGammaTurbulence.m	Rytov/Gamma-Gamma fading.
receiverModel.m	Photodetection, noise, saturation, sampling, threshold, BER.
jammerModel.m	Constant optical jammer.
randomJammerModel.m	Bernoulli intermittent jammer.
test01_perfectChannel.m	Ideal-channel validation.
test02_receiverNoise.m	Noise sensitivity validation.
EXP_BASE_01_distanceSweep.m	Distance study.
EXP_BASE_02_turbulenceSweep.m	Turbulence study.
EXP_BASE_03_pointingSweep.m	Pointing study.
EXP_BASE_04_channelRealization.m	Channel realization study.
EXP_05_constantJammer.m	Controlled constant-jammer comparison.
EXP_06_jammerPowerSweep.m	Constant jammer sweep.
EXP_07_randomJammerValidation.m	Random activity validation.
EXP_08_randomJammerSweep.m	Random activity sweep.
EXP_09_constantJammerBlindingAttack.m	Constant blinding sweep.
EXP_10_randomJammerBlindingAttack.m	Random blinding sweep.
EXP_11_fairJammerComparison.m	Equal-average-power comparison.

Appendix B. Key Equations

Physical Quantity	Governing Equation
Bit duration	$T_b = 1 / R_b$
Sampling frequency	$F_s = R_b \cdot \text{samplesPerBit}$
OOK	$P_{tx,k} = P_t \cdot b_k$
Beam waist	$w_0 = \lambda / (\pi\theta)$
Rayleigh range	$z_R = \pi w_0^2 / \lambda$
Beam radius	$w_{Rx} = w_0 \sqrt{1 + (L / z_R)^2}$
Geometric gain	$H_{geo} = 1 - \exp(-2R_{Rx}^2 / w_{Rx}^2)$
Atmospheric gain	$H_{atm} = 10^{-\alpha L_{km}/10}$
Deterministic gain	$H_{total} = H_{geo} \cdot H_{atm}$
Pointing gain	$H_{point} = \exp(-2r_e^2 / w_{Rx}^2)$
Rytov variance	$\sigma_R^2 = 1.23 \cdot C_n^2 \cdot k^{7/6} \cdot L^{11/6}$
Turbulence	$H_{turb} = X \cdot Y$
Legitimate signal	$P_{rx,legit} = P_{tx} \cdot H_{total} \cdot H_{point} \cdot H_{turb}$
Attacked signal	$P_{rx,total} = P_{rx,legit} + P_{jammer}$
Photocurrent	$I = R \cdot P_{rx,total} + n$
Saturation	$I_{out} = \min(I, I_{sat})$
Threshold	$I_{th} = R \cdot P_t \cdot H_{total} / 2$
BER	$BER = \text{errors} / N_{bits}$
Random average power	$P_{avg} = p \cdot P_{ON}$
Fair activity	$p = P_{avg} / P_{ON}$

Appendix C. Key Experimental Results

EXP 09

$P_j=0.000$ W $\rightarrow$ BER 0.1177, mainly 1 $\rightarrow$ 0
 $P_j=0.100$ W $\rightarrow$ BER 0.0018, helpful-bias region
 $P_j=0.120$ W $\rightarrow$ BER 0.3026, strong 0 $\rightarrow$ 1 transition
 $P_j\geq 0.150$ W $\rightarrow$ BER $\approx$ 0.5076, false-one/blinding plateau
 $P_j\geq 0.400$ W $\rightarrow$ saturation fraction=1.0 in recorded run

EXP 10

Activity 0.0 $\rightarrow$ BER 0.1193, Sat 0.108
Activity 0.5 $\rightarrow$ BER 0.3180, Sat 0.568
Activity 0.8 $\rightarrow$ BER 0.4306, Sat 0.833
Activity 1.0 $\rightarrow$ BER 0.4990, Sat 1.000

EXP 11

Pavg 0.05 W: Constant BER 0.033, Random BER 0.161
Pavg 0.10 W: Constant BER 0.003, Random BER 0.207
Pavg 0.15 W: Constant BER 0.494, Random BER 0.247
Pavg 0.30 W: Constant BER 0.494, Random BER 0.397
Pavg 0.40 W: Constant BER 0.494, Random BER 0.494

Final project narrative

The project has progressed from first-principles optical communication to an experimentally characterized physical-layer security problem. The baseline explains how legitimate information is transmitted and degraded. The jammer experiments reveal multiple attack regimes rather than a single monotonic failure mechanism. The blinding experiments connect excessive optical power to receiver saturation and information collapse. The fair comparison establishes that temporal attack structure and instantaneous peak power matter even under equal average power budgets. These signatures provide the correct foundation for the next ML phase.